Matching Continued

STSCI/INFO/ILRST 3900: Causal Inference

29 Oct 2026

Learning goals for today

At the end of class, you will be able to:

1. Understand propensity score matching and coarsened exact matching
2. Use matching methods to estimate causal effects

Logistics

- Quiz 3: Nov 5
- Practice quiz review in-class Tuesday
- My office OH moved to Wed

Matching: The big idea

Goal: Estimate the ATT:

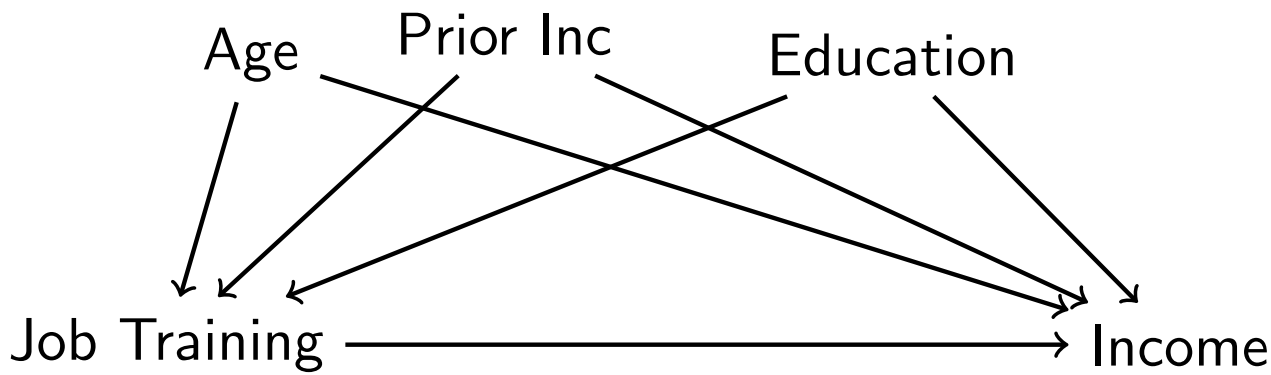
$$\mathrm{E}(Y^{a=1} \mid A = 1) - \mathrm{E}(Y^{a=0} \mid A = 1)$$

Potential Solution:

For each treated individual i, get a set of **untreated** individuals, $\mathcal{M}_i$, and use their observations to estimate $Y_i^{a=0}$

$$\mathrm{E}(Y^{a=0} \mid A = 1) \approx \frac{1}{n_t} \sum_{i:A_i=1} Y_i^{a=0} \approx \frac{1}{n_t} \sum_{i:A_i=1} \left(\underbrace{\frac{1}{|\mathcal{M}_i|} \sum_{i \text{ in } \mathcal{M}_i} Y_i}_{\text{Avg over } \mathcal{M}_i} \right)$$

Motivating example



- L denotes the set of covariates we want to condition on
- ℓ_i denotes the covariates values for individual i

A common distance metric: Exact matching

- Ideally, we find an exact match for each treated unit

$$d(i,j) = \begin{cases} 0 & \text{if } \ell_i = \ell_j \\ \infty & \text{if } \ell_i \neq \ell_j \end{cases}$$

Because of the curse of dimensionality, this often leads to **no matches at all**

A common distance metric: Coarsened exact matching[1]

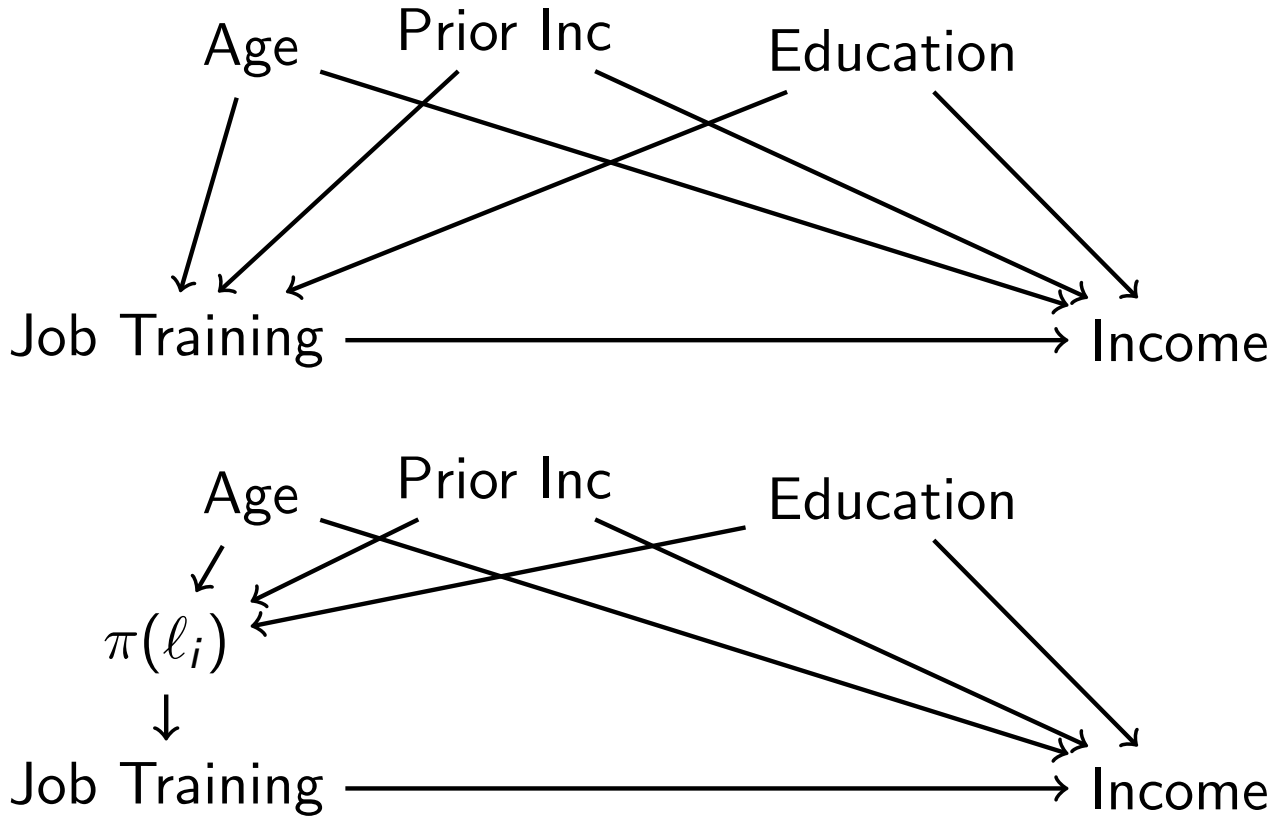
- Define $\tilde{L}$ to be a coarsened version of L
  - Example: Age 15-20, 20-25, 25-30, etc
- Match exactly on $\tilde{L}$

$$d(i,j) = \begin{cases} 0 & \text{if } \tilde{\ell}_i = \tilde{\ell}_j \\ \infty & \text{if } \tilde{\ell}_i \neq \tilde{\ell}_j \end{cases}$$

- Benefit: Very transparent
- Benefit: Directly targets balance in L
- Drawback: Need to define the "buckets"
- Drawback: Still may not find a good match for all individuals

[1] Iacus, S. M., King, G., & Porro, G. (2012). Causal inference without balance checking: Coarsened exact matching. Political Analysis, 20(1), 1-24.

A common distance metric: Propensity scores



Suppose L only affects A through a probability of treatment

$$\pi_i = \pi(\ell_i) = P(A_i = 1 \mid L = \ell_i)$$

Conditional exchangeability holds given $\pi(\ell_i)$

Why propensity scores are nice

- Can match on propensity scores directly instead of L
  - Easy to reason about
  - Can directly visualize the univariate matches
- Intuitive: Prioritizes covariates that predict treatment
- Mathematical guarantees on average
  - If our DAG is correct
  - If our matches are good
  - We should **on average** get a matched group which looks like the the treatment group

$$P(L \mid \pi_i, A_i = 1) = P(L \mid \pi_i, A_i = 0)$$

Multivariate distances: Recap

When matching on multivariate L, you have to define the distance between each pair of confounder values $\vec{\ell}_j$ and $\vec{\ell}_i$

- Manhattan distance
- Euclidean distanace
- Mahalanobis distance
- Coarsened exact distance
- Propensity score distance

There is no right answer! Depends on the setting.

- Propensity scores are most popular
- Sometimes they are substantively meaningful
- Balance only occurs on average

Evaluate the matched sets

Whatever method, you should check that it worked

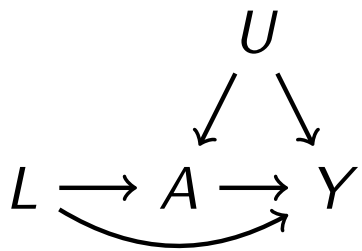
- Compare means of L (propensity scores) across groups
- Possibly compare interaction cells; e.g., race $\times$ age
- Visually assess distribution
- Lack of overlap may indicate violation of positivity assumption

Matching: A word of warning[2]

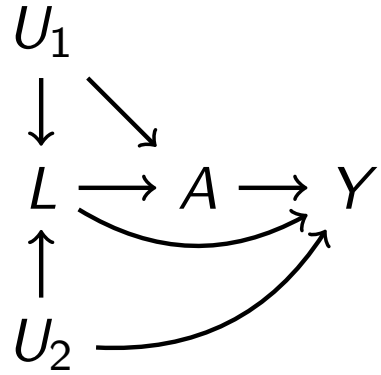
Matching works!



No help!



No help!



Matching is an estimation strategy.
It does not solve identification problems.

[2]Sekhon, J. S. (2009). Opiates for the matches: Matching methods for causal inference. Annual Review of Political Science, 12(1), 487-508.

Estimating a causal effect

- If we've matched everything well, we can compare the means
  - Treated group (with a match)
  - Matched control group
- We can be extra careful by combining regression + matching
  - If everything is perfect, both should be fine on their own
  - Combining can reduce bias
  - Reduces model sensitivity[3]

[3] On the statistical role of inexact matching in observational studies. Guo and Rothenhäusler (2023)

Code example

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